TAKE-HOME ASSIGNMENT

Lumbar Spine Segmentation from CT

Preprocessing · 3D segmentation · Vertebral body centroids · Cross-dataset evaluation

EFFORT	DEADLINE	MAIN DELIVERABLE
About one working day	**5 days after receipt**	**Two-page PDF report**

1 Overview

You will build a small pipeline for lumbar CT: load and standardise data from two different datasets, segment the vertebrae L1 to L5, find the vertebral bodies, and evaluate how well your model transfers to a different dataset. Your main deliverable is a two-page report that explains and justifies your methods and results. We care more about sound engineering and clear reasoning than about the highest possible score.

2 Data

Download both datasets yourself:

1. **TotalSegmentator** github.com/wasserth/TotalSegmentator
   Use the official train/test split.
2. **VerSe 2019/2020** github.com/anjany/verse

Target labels: vertebrae **L1, L2, L3, L4, L5**.

The two datasets differ in orientation, field of view, label convention, scanners and pathology. Handling this is part of the task.

3 Tasks

The assignment has four parts (A–D).

3.1 Part A: Preprocessing

Write a loader that works on **both** datasets. For every case, it:

- converts image and labels to one canonical orientation,
- resamples to a common voxel spacing of your choice,
- normalises the intensities.

Report every header anomaly you find. For each one, say how you detected it and how you handled it.

3.2 Part B: Segmentation

Train a 3D segmentation model yourself on TotalSegmentator data. It should segment the vertebrae L1, L2, L3, L4 and L5, each as a separate class.

Any framework is allowed. A small model or a short training run is fine. We care about the pipeline and the analysis, not the highest score.

> **Not allowed:** pretrained spine models (e.g. TotalSegmentator, VerSe models).

3.3 Part C: Postprocessing

For each vertebra (L1–L5), compute the centroid of the **vertebral body only**, without the posterior elements (pedicles, laminae, spinous and transverse processes). Neither dataset has a body label, so how you separate the body is up to you. Explain and justify your method.

Output one JSON file per case:

```
{
  "case_id": "s0001",
  "coordinate_system": "RAS",
  "unit": "mm",
  "centroids": {
    "L2": [12.3, -45.6, 78.9],
    "L3": [12.1, -44.8, 49.2]
  }
}
```

The coordinates are world coordinates in mm. State the convention (RAS or LPS) in the file.

> **Optional:** VerSe provides centroid annotations. If you use them for validation, explain what exactly they describe and how they relate to your vertebral body centroids.

3.4 Part D: Cross-Dataset Evaluation

Evaluate the model you trained in Part B on:

1. the TotalSegmentator test split (in-domain),
2. VerSe (out-of-domain).

Harmonise the label conventions of both datasets and document your mapping, including transitional vertebrae (e.g. L6).

Report the Dice score per vertebra (L1–L5) on both test sets, at minimum. Why are there differences? Did you notice anything odd?

4 Deliverables

4.1 Report (main deliverable)

A two-page PDF. Explain and justify your methodology and your results:

- preprocessing and the header anomalies you found (Part A),
- model and training setup, and why you chose them (Part B),
- your method for vertebral body centroids (Part C),
- results on both test sets with Dice per vertebra, and your analysis of the differences (Part D),
- what you would do next with more resources, and what you would deliberately *not* do, and why.

Two pages maximum. Tables and figures count toward the limit.

4.2 Code

Your code as a single zip file, including the model weights and all files needed for inference. Please do not include the training data.

4.3 Inference script

An inference script in the zip. It must run on a fresh machine with a single command, for example:

```
./run_inference.sh <input_dir> <output_dir>
```

Document the requirements (OS, GPU yes/no, Docker or not).

5 Hardware Tips

You do not need your own GPU. These options are free or cheap:

- **Free notebooks:** Kaggle Notebooks (weekly free GPU quota) and Google Colab (free tier with a T4 GPU, sessions time out). Save checkpoints regularly so a disconnect does not cost you the run.
- **Rented cloud GPUs:** providers such as RunPod, Vast.ai or Lambda rent GPUs by the hour, often for well under 1 USD/h. A few hours are enough for this task. Stop the instance when you are done.
- **Free credits:** several cloud providers offer starter credits for new accounts.

Make training cheap, too:

- Crop to the lumbar region (the labels tell you where it is) and resample to a coarse spacing, e.g. 1.5–2 mm.
- Train on patches with mixed precision, and start with a subset of the training cases.
- Do all preprocessing on your own machine (CPU) and upload only the prepared data.
- Inference on a few cases also runs on a CPU.

6 Rules

- AI coding assistants are allowed. In a **30-minute code review** afterwards, you must be able to explain your algorithm and your design decisions.
- If you run short on time, prioritise: a working end-to-end pipeline with honest reporting is better than one polished part.

Data protection is your responsibility. You alone are responsible for complying with the GDPR, all other applicable data protection rules, and the licences and terms of use of the datasets. This includes how and where you download, store and process the data, e.g. on cloud services.